# Module 3 Assignment: Target Triage with Protein Language Models

**Disease: Lung Adenocarcinoma (LUAD)**

**1. Re-ranking**

My Module 3 results partly agreed with my Module 2 rankings, but some targets changed after including drug tractability scores.

COL6A6 moved from rank 3 to rank 1, and SCN7A moved from rank 5 to rank 3. The biggest change was ZNF536, which dropped from rank 2 to rank 7. Although ZNF536 had a high Module 2 score (0.988), its small-molecule score was only 0.003, and its antibody score was almost zero. This suggests that it does not strongly resemble previously drugged proteins. My Module 2 results also showed opposite RNA and protein changes for ZNF536.

I selected **w = 0.6**, giving 60% weight to Module 2 biological evidence and 40% to tractability. I chose this because a protein should first have strong evidence of disease involvement before we consider how easily it could be targeted by drugs.

**2. Modality Comparison**

I selected SCN7A because its small-molecule score was 0.645, while its antibody score was approximately 0.000.

SCN7A belongs to the sodium-channel family and is located in the cell membrane. Proteins in this family can have regions that small molecules interact with to change their activity. Antibodies usually need accessible regions on the outside of a protein to bind effectively.

This may explain why SCN7A received a higher small-molecule score. However, we cannot confirm whether SCN7A has a suitable binding pocket from these scores alone. Structural analysis is needed to investigate this.

**3. Model Validation**

I evaluated the model using random-split and family-split validation.

| Validation | AUROC | Average Precision |
|---|---|---|
| Random split | 0.844 | 0.724 |
| Family split | 0.726 | 0.537 |

The chance level for average precision was 0.334.

The family-split results were lower because this method keeps related protein families separate during training and testing.

In random splitting, similar proteins may appear in both datasets, making the model perform better than it would on unfamiliar protein families.

I would report the **family-split AUROC of 0.726 and average precision of 0.537** because they provide a more realistic estimate of model performance. The results are better than chance, but they are not strong enough to confirm druggability.

I also removed 2 overlapping targets from the training panel to prevent data leakage, leaving 1,199 proteins for training.

**4. Anchor Variant Interpretation**

I analyzed four anchor variants using the protein language model.

| Gene | Variant | Score | Percentile |
|---|---|---|---|
| APOE | R176C (rs7412, ε2) | -8.56 | 11.8 |
| APOE | C130R (rs429358, ε4) | 7.21 | 100.0 |
| TREM2 | R47H | -5.16 | 47.7 |
| TP53 | R175H | -5.97 | 19.3 |

I used UniProt numbering for APOE, including its 18-amino-acid signal peptide. The corresponding mature-protein variants are R158C and C112R.

APOE R176C had the most negative score, suggesting that the model considers this substitution highly unusual. APOE C130R had a positive score, meaning the model considered the changed amino acid more likely than the original one.

TP53 R175H also had a strongly negative score, while TREM2 R47H was closer to the middle of its protein's score distribution.

However, **a negative score does not automatically mean a mutation is pathogenic**. It only indicates how unusual the change appears based on learned protein sequences. These models cannot directly detect noncoding mutations, regulatory changes, or RNA splicing effects.

Therefore, experimental and clinical evidence is needed to understand whether these variants actually contribute to disease.

**5. Limitations and Next Step**

Protein language models can identify similarities between proteins, but sequence information alone cannot confirm the presence of a suitable drug-binding pocket. It also cannot tell us whether a drug will bind effectively or improve the disease.

For Week 4, I would select **SCN7A and MASP1** for structure-based pocket assessment.

**SCN7A:** I selected SCN7A because it had the highest small-molecule tractability score (0.645) among my targets. Since it belongs to the sodium-channel family, I would like to investigate whether its structure contains an accessible binding region for small molecules.

**MASP1:** I selected MASP1 because it had relatively high small-molecule (0.605) and antibody (0.550) scores. MASP1 is a serine protease, so I would like to investigate its enzyme active site and check whether it could be targeted by drugs.

However, my Module 2 results showed decreased RNA and protein levels for both targets. Therefore, we need to understand whether changing their activity would actually benefit LUAD patients.

These two targets would be my next choices for structural analysis, but further biological validation is still necessary.

**Unmapped Genes**

All 15 LUAD targets were successfully mapped to the human proteome dataset. There were **0 unmapped genes**, so no alternative evidence was needed for missing protein mappings.